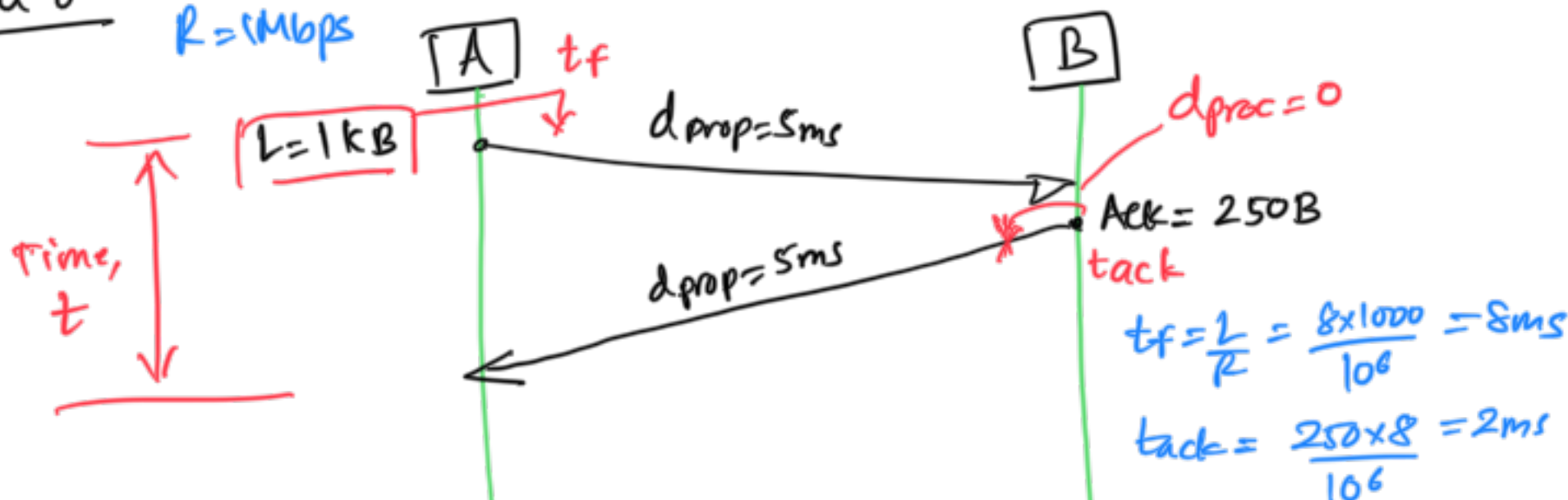


Communication models

* Bandwidth Delay Product (BDP)

Q.6.

$R = 1 \text{ Mbps}$



$$\text{Throughput} = \frac{L}{t}$$

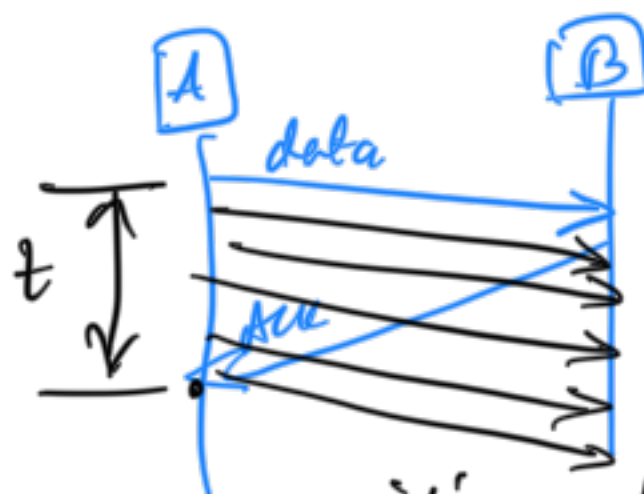
$$= \frac{8 \times 1000}{20 \times 10^{-3}}$$

$$= 400 \text{ Kbps}$$

$$t = t_f + d_{\text{prop}} * 2 + t_{\text{ack}}$$

$$t = 8 + 10 + 2$$

$$t = 20 \text{ ms}$$



Amount of data sent in ' t ' sec determine the data size to fill the pipe

$$R = 1 \text{ Mbps}$$

$$t = 20 \text{ ms}$$

$$1 \text{ Mb} \dots 1 \text{ s}$$

$$\# \text{ bits to fill the pipe} = R * t$$

$$= 20,000 \text{ bits} \Rightarrow 205 \text{ packets}$$

Bandwidth-delay product

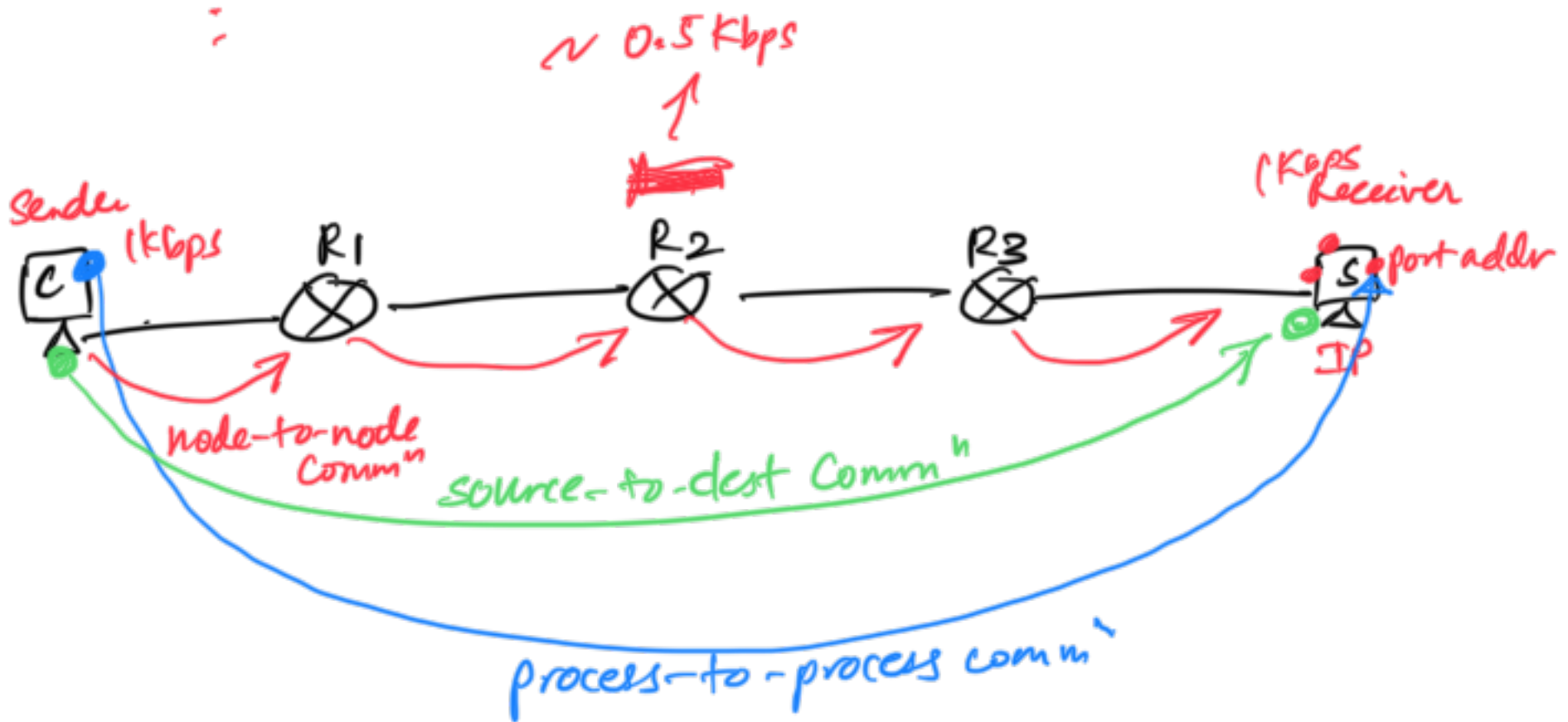
↓
Bottleneck
BW

- * BDP is used to determine the number of bytes the sender should continuously send before waiting for an ACK to fill the pipe betⁿ the sender & receiver. This provides the max. possible throughput betⁿ the sender & receiver.
- * This also signifies the amount of data in flight before you hear about packet losses, etc
- * We will revisit this ^{BDP} concept when we study TCP protocol.

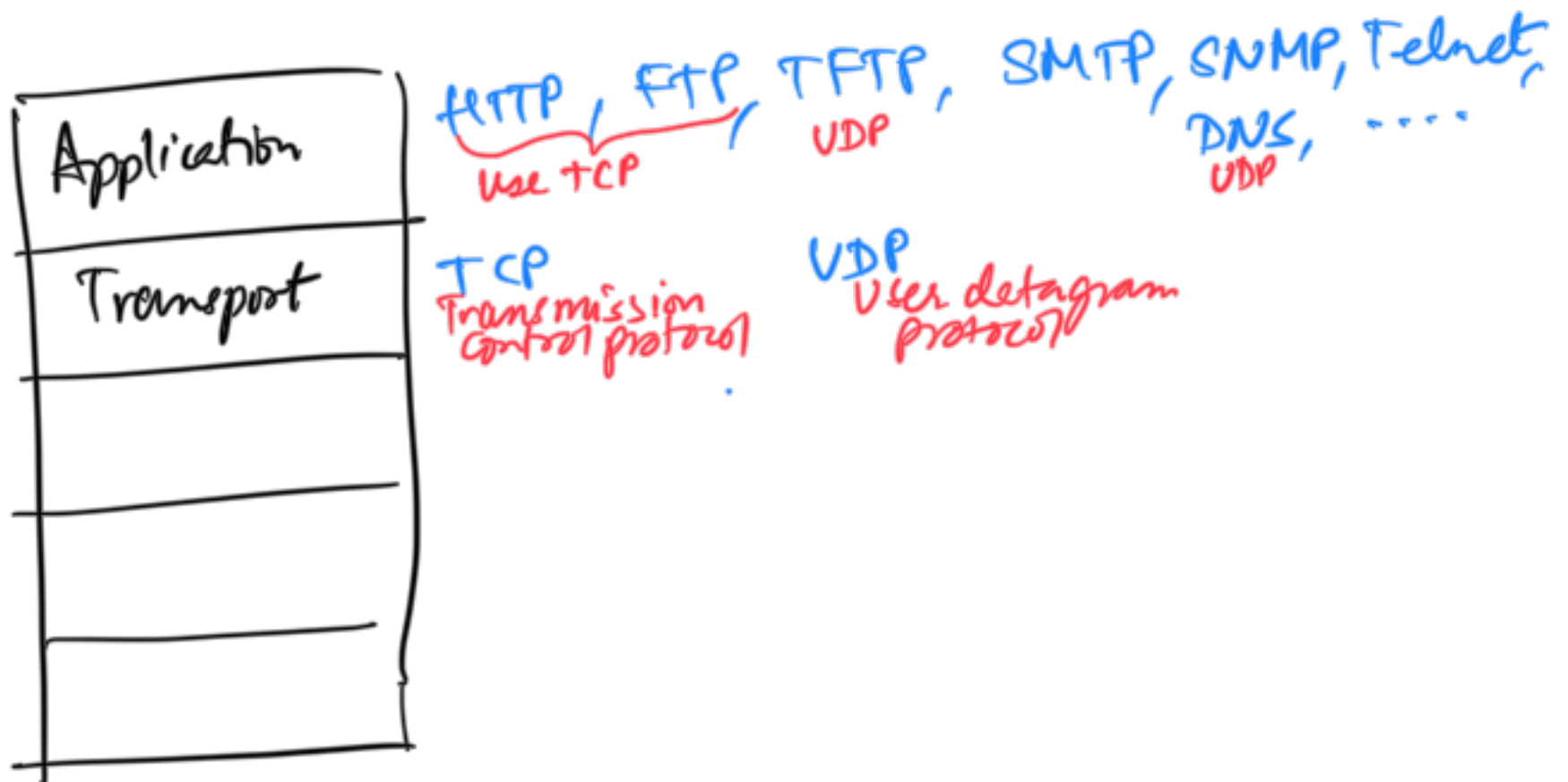
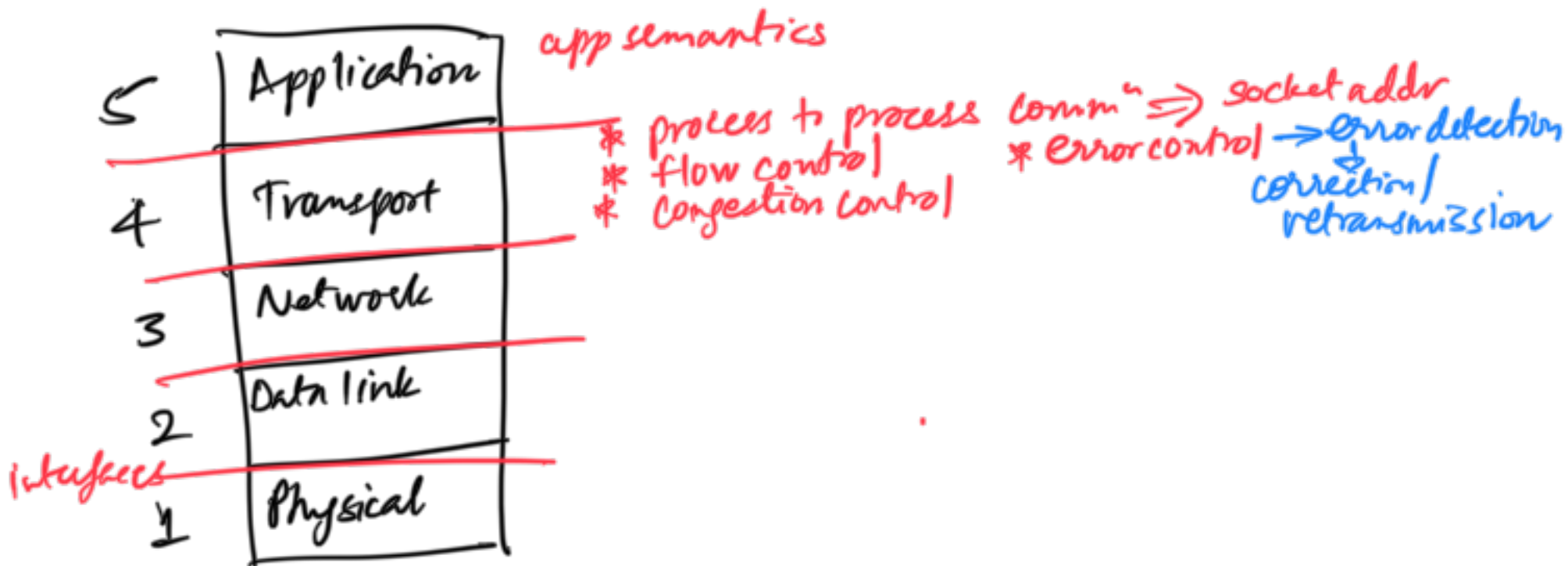
communication models

Challenges

- * Packet losses $\Rightarrow$ how to recover?
- * Out of order packet delivery
- * Congestion
- * Failure ... Nodes in the Internet
- * Addressing.



Internet model



TCP	UDP
* <u>Connⁿ</u> oriented ↳ Call setup / termⁿ	* <u>Connⁿ</u> less ↳ X Call setup
* Reliable ↳ recovery	* Unreliable ↳ ≠ recovery mech.